개인 보고서

한경식

PDP 측정 데이터

디스플레이 종류	PDP	명암	표준(80)	밝기	표준(50)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	66.60	66.48	1.8235	1.8227		
25	245	66.15	66.03	1.8205	1.8197		
24	235	65.03	64.91	1.8131	1.8123		
23	225	64.64	64.52	1.8105	1.8097		
22	215	61.64	61.52	1.7899	1.7890		
21	205	60.87	60.75	1.7844	1.7835		
20	195	59.14	59.02	1.7719	1.7710		
19	185	57.62	57.5	1.7605	1.7597		
18	175	54.40	54.28	1.7356	1.7346		
17	165	51.36	51.24	1.7106	1.7096		
16	155	48.84	48.72	1.6888	1.6877		
15	145	44.06	43.94	1.6440	1.6429		
14	135	41.09	40.97	1.6137	1.6125		
13	125	38.40	38.28	1.5843	1.5830		
12	115	33.16	33.04	1.5205	1.5190		
11	105	27.41	27.29	1.4379	1.4360		
10	95	21.14	21.02	1.3250	1.3226		
9	85	16.55	16.43	1.2189	1.2156		
8	75	12.55	12.43	1.0986	1.0945		
7	65	9.42	9.3	0.9741	0.9685		
6	55	6.31	6.19	0.7997	0.7917		
5	45	3.86	3.74	0.5860	0.5729		
4	35	2.24	2.12	0.3493	0.3263		
3	25	1.62	1.5	0.2095	0.1761		
2	15	1.04	0.92	0.0149	-0.0362		
1	5	0.12	0	-0.9208	#NUM!		
0	0	0.12	= L0	————	————————		

PDP 측정 데이터

명암 변화

디스플레이 종류	PDP	명암	100(+20)	밝기	표준(50)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	71.74	71.62	1.8557	1.8550		
25	245	71.12	71	1.8520	1.8513		
24	235	69.91	69.79	1.8445	1.8438		
23	225	69.83	69.71	1.8441	1.8433		
22	215	67.42	67.3	1.8288	1.8280		
21	205	66.46	66.34	1.8226	1.8218		
20	195	64.87	64.75	1.8120	1.8112		
19	185	63.34	63.22	1.8017	1.8009		
18	175	61.06	60.94	1.7857	1.7849		
17	165	58.54	58.42	1.7674	1.7666		
16	155	57.94	57.82	1.7629	1.7621		
15	145	54.95	54.83	1.7400	1.7390		
14	135	50.42	50.3	1.7026	1.7016		
13	125	47.29	47.17	1.6748	1.6737		
12	115	43.25	42.13	1.6360	1.6246		
11	105	38.33	38.21	1.5835	1.5822		
10	95	31.75	31.63	1.5017	1.5001		
9	85	25.54	25.42	1.4072	1.4052		
8	75	18.63	18.51	1.2703	1.2674		
7	65	14.14	14.02	1.1504	1.1467		
6	55	9.65	9.54	0.9844	0.9795		
5	45	6.15	6.03	0.7890	0.7803		
4	35	3.47	3.35	0.5403	0.5250		
3	25	1.45	1.33	0.1626	0.1239		
2	15	0.42	0.3	-0.3809	-0.5229		
1	5	0.12	0	-0.9208	#NUM!		
0	0	0.12	= L0	————	—————————		

디스플레이 종류	PDP	명암	60(-20)	밝기	표준(50)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	60.5	60.4	1.7818	1.7810		
25	245	58.77	58.67	1.7692	1.7684		
24	235	57.94	57.84	1.7630	1.7622		
23	225	56.36	56.26	1.7510	1.7502		
22	215	54.26	54.16	1.7345	1.7337		
21	205	52.96	52.86	1.7239	1.7231		
20	195	52.25	52.15	1.7181	1.7173		
19	185	49.15	49.05	1.6915	1.6906		
18	175	47.24	47.14	1.6743	1.6734		
17	165	43.62	43.52	1.6397	1.6387		
16	155	36.83	36.73	1.5662	1.5650		
15	145	34.2	34.1	1.5340	1.5328		
14	135	28.93	28.83	1.4613	1.4598		
13	125	24.55	24.45	1.3901	1.3883		
12	115	19.92	19.82	1.2993	1.2971		
11	105	15.77	15.67	1.1978	1.1951		
10	95	12.9	12.8	1.1106	1.1072		
9	85	9.84	9.74	0.9930	0.9886		
8	75	7.87	7.77	0.8960	0.8904		
7	65	5.13	5.03	0.7101	0.7016		
6	55	3.55	3.45	0.5502	0.5378		
5	45	2.28	2.18	0.3579	0.3385		
4	35	2.24	2.14	0.3502	0.3304		
3	25	0.83	0.73	-0.0809	-0.1367		
2	15	0.44	0.34	-0.3565	-0.4685		
1	5	0.11	0.01	-0.9586	-2.0000		
0	0	0.1	= L0	————	—————————		

PDP 측정 데이터

밝기 변화

디스플레이 종류	PDP	명암	표준(80)	밝기	70(+20)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	67.77	67.27	1.8310	1.8278		
25	245	67.62	67.12	1.8301	1.8269		
24	235	67.04	66.54	1.8263	1.8231		
23	225	66.25	65.75	1.8212	1.8179		
22	215	66.17	65.67	1.8207	1.8174		
21	205	64.74	64.24	1.8112	1.8078		
20	195	63.5	63	1.8028	1.7993		
19	185	60.03	59.53	1.7784	1.7747		
18	175	58.75	58.25	1.7690	1.7653		
17	165	57.64	57.14	1.7607	1.7569		
16	155	53.43	52.93	1.7278	1.7237		
15	145	52.11	51.61	1.7169	1.7127		
14	135	48.26	47.76	1.6836	1.6791		
13	125	44.46	43.96	1.6480	1.6431		
12	115	41.27	40.77	1.6156	1.6103		
11	105	36.36	35.86	1.5606	1.5546		
10	95	33.31	32.81	1.5226	1.5160		
9	85	27.63	27.13	1.4414	1.4334		
8	75	21.4	20.9	1.3304	1.3201		
7	65	16.65	16.15	1.2214	1.2082		
6	55	12.84	12.34	1.1086	1.0913		
5	45	9.49	8.99	0.9773	0.9538		
4	35	6.55	6.05	0.8162	0.7818		
3	25	4.05	3.55	0.6075	0.5502		
2	15	2.7	2.2	0.4314	0.3424		
1	5	0.83	0.33	-0.0809	-0.4815		
0	0	0.5	= L0	————	————————		

디스플레이 종류	PDP	명암	표준(80)	밝기	30(-20)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	62.87	62.77	1.7984	1.7978		
25	245	61.54	61.44	1.7892	1.7885		
24	235	60.15	60.05	1.7792	1.7785		
23	225	59.44	59.34	1.7741	1.7733		
22	215	57.8	57.7	1.7619	1.7612		
21	205	56.55	56.45	1.7524	1.7517		
20	195	52.93	52.83	1.7237	1.7229		
19	185	50.52	50.42	1.7035	1.7026		
18	175	47.88	47.78	1.6802	1.6792		
17	165	43.28	43.18	1.6363	1.6353		
16	155	40.2	40.1	1.6042	1.6031		
15	145	37.07	36.97	1.5690	1.5678		
14	135	31.84	31.74	1.5030	1.5016		
13	125	26.51	26.41	1.4234	1.4218		
12	115	20.62	20.52	1.3143	1.3122		
11	105	15.8	15.7	1.1987	1.1959		
10	95	11.66	11.56	1.0667	1.0630		
9	85	8.74	8.64	0.9415	0.9365		
8	75	5.63	5.53	0.7505	0.7427		
7	65	3.53	3.43	0.5478	0.5353		
6	55	0.67	0.57	-0.1739	-0.2441		
5	45	0.16	0.06	-0.7959	-1.2218		
4	35	0.11	0.01	-0.9586	-2.0000		
3	25	0.11	0.01	-0.9586	-2.0000		
2	15	0.11	0.01	-0.9586	-2.0000		
1	5	0.11	0.01	-0.9586	-2.0000		
0	0	0.1	= L0	————	————————		

LCD 측정 데이터

디스플레이 종류	LCD	명암	표준(57)	밝기	표준(43)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	336.25	335.75	2.5267	2.5260		
25	245	304.64	304.14	2.4838	2.4831		
24	235	276.17	275.67	2.4412	2.4404		
23	225	250.77	250.27	2.3993	2.3984		
22	215	223.03	222.53	2.3484	2.3474		
21	205	199.55	199.05	2.3001	2.2990		
20	195	177.23	176.73	2.2485	2.2473		
19	185	156.51	156.01	2.1945	2.1932		
18	175	177.2	176.7	2.2485	2.2472		
17	165	156.55	156.05	2.1947	2.1933		
16	155	104.84	104.34	2.0205	2.0185		
15	145	91.37	90.87	1.9608	1.9584		
14	135	76.88	76.38	1.8858	1.8830		
13	125	65.35	64.85	1.8152	1.8119		
12	115	52.73	52.23	1.7221	1.7179		
11	105	41.77	41.27	1.6209	1.6156		
10	95	32.08	31.58	1.5062	1.4994		
9	85	24.34	23.84	1.3863	1.3773		
8	75	17.91	17.41	1.2531	1.2408		
7	65	11.84	11.34	1.0734	1.0546		
6	55	7.44	6.94	0.8716	0.8414		
5	45	4.43	3.93	0.6464	0.5944		
4	35	2.21	1.71	0.3444	0.2330		
3	25	1.11	0.61	0.0453	-0.2147		
2	15	0.63	0.13	-0.2007	-0.8861		
1	5	0.52	0.02	-0.2840	-1.6990		
0	0	0.5	= L0	————	————————		

LCD 측정 데이터

명암 변화

디스플레이 종류	LCD	명암	77(+20)	밝기	표준(43)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	525.73	525.21	2.7208	2.7203		
25	245	497.55	497.03	2.6968	2.6964		
24	235	460.52	460	2.6632	2.6628		
23	225	415.44	414.92	2.6185	2.6180		
22	215	268.87	268.35	2.4295	2.4287		
21	205	368.88	368.36	2.5669	2.5663		
20	195	292.04	291.52	2.4654	2.4647		
19	185	259.54	259.02	2.4142	2.4133		
18	175	225.37	224.85	2.3529	2.3519		
17	165	195.32	194.8	2.2907	2.2896		
16	155	167.22	166.7	2.2233	2.2219		
15	145	143.11	142.59	2.1557	2.1541		
14	135	121.8	121.28	2.0856	2.0838		
13	125	102.43	101.91	2.0104	2.0082		
12	115	85.66	85.14	1.9328	1.9301		
11	105	68.73	68.21	1.8371	1.8338		
10	95	54.03	53.51	1.7326	1.7284		
9	85	40.66	40.14	1.6092	1.6036		
8	75	28.85	29.33	1.4601	1.4673		
7	65	20.07	19.55	1.3025	1.2911		
6	55	12.05	11.53	1.0810	1.0618		
5	45	6.44	5.92	0.8089	0.7723		
4	35	3.12	2.6	0.4942	0.4150		
3	25	3.1	2.58	0.4914	0.4116		
2	15	0.64	0.12	-0.1938	-0.9208		
1	5	0.55	0.03	-0.2596	-1.5229		
0	0	0.52	= L0	————	—————————		

디스플레이 종류	LCD	명암	37(-20)	밝기	표준(43)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	189.03	188.5	2.2765	2.2753		
25	245	171.43	170.6	2.2341	2.2320		
24	235	156.54	156.01	2.1946	2.1932		
23	225	142.37	141.84	2.1534	2.1518		
22	215	127.87	123.54	2.1068	2.0918		
21	205	115.94	115.41	2.0642	2.0622		
20	195	105.55	105.02	2.0235	2.0213		
19	185	95.25	96.72	1.9789	1.9855		
18	175	84.12	83.59	1.9249	1.9222		
17	165	74.66	74.13	1.8731	1.8700		
16	155	64.09	63.56	1.8068	1.8032		
15	145	55.24	54.71	1.7423	1.7381		
14	135	46.53	46	1.6677	1.6628		
13	125	37.88	37.35	1.5784	1.5723		
12	115	30.81	30.28	1.4887	1.4812		
11	105	24.96	24.43	1.3972	1.3879		
10	95	19.15	18.62	1.2822	1.2700		
9	85	14.66	14.13	1.1661	1.1501		
8	75	10.82	10.29	1.0342	1.0124		
7	65	7.35	6.82	0.8663	0.8338		
6	55	4.95	4.42	0.6946	0.6454		
5	45	3.01	2.48	0.4786	0.3945		
4	35	1.71	1.18	0.2330	0.0719		
3	25	1.04	0.51	0.0170	-0.2924		
2	15	0.68	0.15	-0.1675	-0.8239		
1	5	0.55	0.02	-0.2596	-1.6990		
0	0	0.53	= L0	————	—————————		

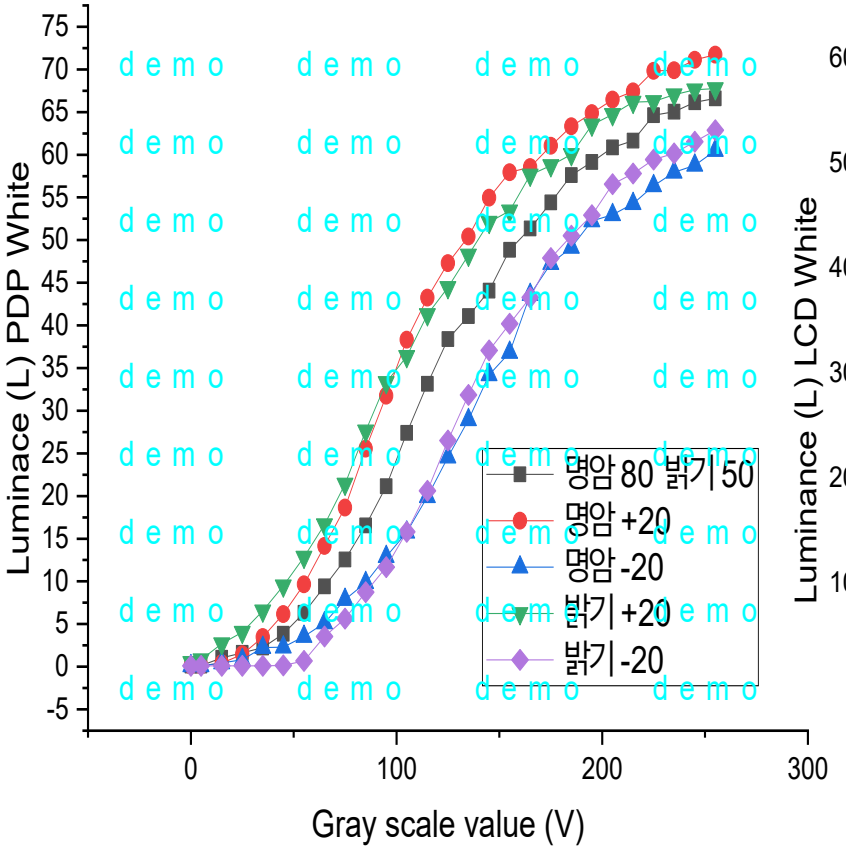
LCD 측정 데이터

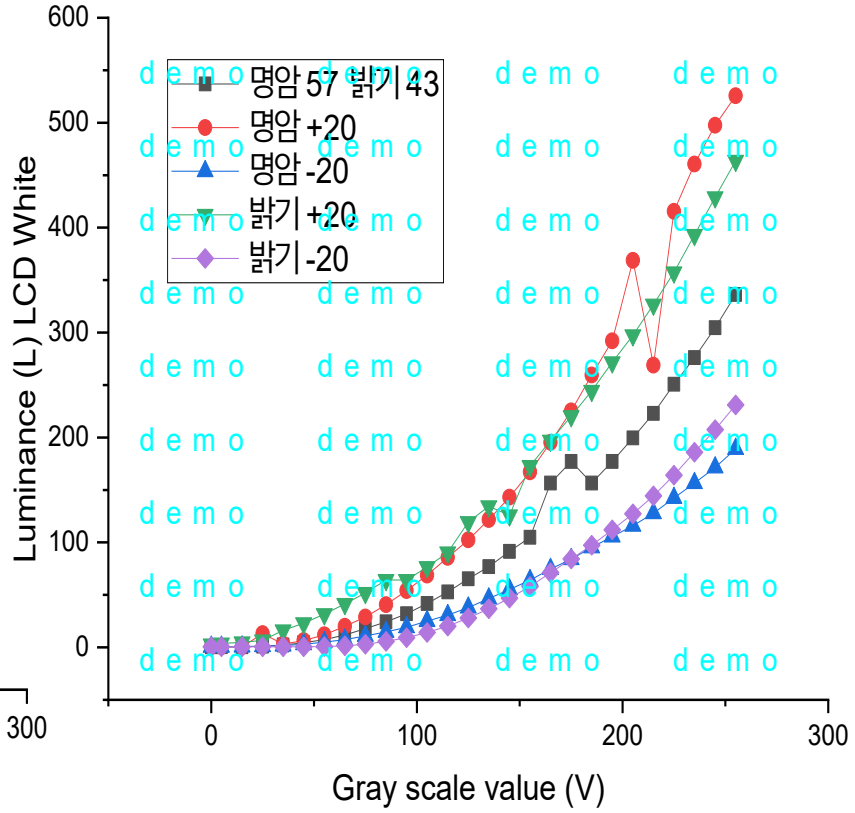
밝기 변화

디스플레이 종류	LCD	명암	표준(57)	밝기	63(+20)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	463.72	460.89	2.6663	2.6636		
25	245	428.98	426.15	2.6324	2.6296		
24	235	393.12	390.29	2.5945	2.5914		
23	225	357.34	354.29	2.5531	2.5494		
22	215	326.65	323.82	2.5141	2.5103		
21	205	297.43	294.6	2.4734	2.4692		
20	195	271.68	268.85	2.4341	2.4295		
19	185	244.5	241.67	2.3883	2.3832		
18	175	220.31	217.48	2.3430	2.3374		
17	165	197.06	194.23	2.2946	2.2883		
16	155	172.96	170.09	2.2379	2.2307		
15	145	125.92	123.09	2.1001	2.0902		
14	135	134.47	131.64	2.1286	2.1194		
13	125	119.35	116.52	2.0768	2.0664		
12	115	90.53	87.7	1.9568	1.9430		
11	105	76.52	73.69	1.8838	1.8674		
10	95	64.27	61.44	1.8080	1.7885		
9	85	64.27	61.44	1.8080	1.7885		
8	75	51.86	49.03	1.7148	1.6905		
7	65	41.48	38.65	1.6178	1.5871		
6	55	31.6	28.77	1.4997	1.4589		
5	45	22.93	20.1	1.3604	1.3032		
4	35	16.15	13.32	1.2082	1.1245		
3	25	6.61	4.56	0.8202	0.6590		
2	15	4.85	2.05	0.6857	0.3118		
1	5	3.66	0.83	0.5635	-0.0809		
0	0	2.83	= L0	————	—————————		

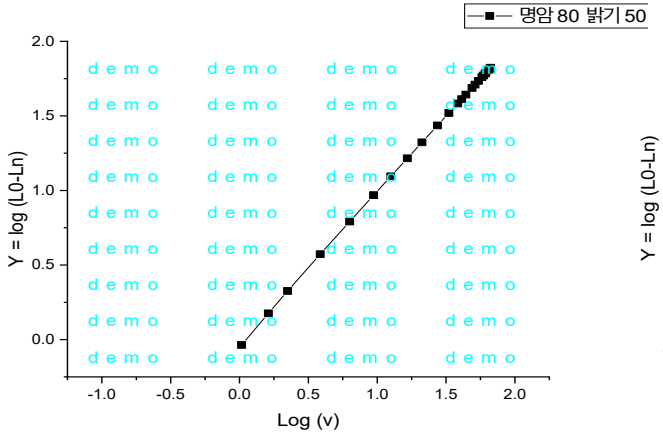
디스플레이 종류	LCD	명암	표준(57)	밝기	23(-20)	Aging Time	10m
측정 데이터 [cd/m^2]							
Drive Level	Gray Level	평균 휘도	L-L0	logV	log(L-L0)		
26	255	231.11	230.56	2.3638	2.3628		
25	245	207.43	206.88	2.3169	2.3157		
24	235	186.07	185.52	2.2697	2.2684		
23	225	164.17	163.62	2.2153	2.2138		
22	215	144.38	143.83	2.1595	2.1578		
21	205	127.31	126.76	2.1049	2.1030		
20	195	112.13	111.58	2.0497	2.0476		
19	185	97.3	96.75	1.9881	1.9857		
18	175	84.26	83.74	1.9256	1.9229		
17	165	71.29	70.74	1.8530	1.8497		
16	155	57.99	57.44	1.7634	1.7592		
15	145	46.92	46.37	1.6714	1.6662		
14	135	36.73	36.18	1.5650	1.5585		
13	125	27.95	27.4	1.4464	1.4378		
12	115	20.35	19.8	1.3086	1.2967		
11	105	14.23	13.67	1.1532	1.1358		
10	95	9.31	8.76	0.9689	0.9425		
9	85	5.74	5.19	0.7589	0.7152		
8	75	2.95	2.4	0.4698	0.3802		
7	65	1.54	0.99	0.1875	-0.0044		
6	55	0.77	0.22	-0.1135	-0.6576		
5	45	0.56	0.01	-0.2518	-2.0000		
4	35	0.55	0	-0.2596	#NUM!		
3	25	0.55	0	-0.2596	#NUM!		
2	15	0.55	0	-0.2596	#NUM!		
1	5	0.55	0	-0.2596	#NUM!		
0	0	0.55	= L0	————	—————————		

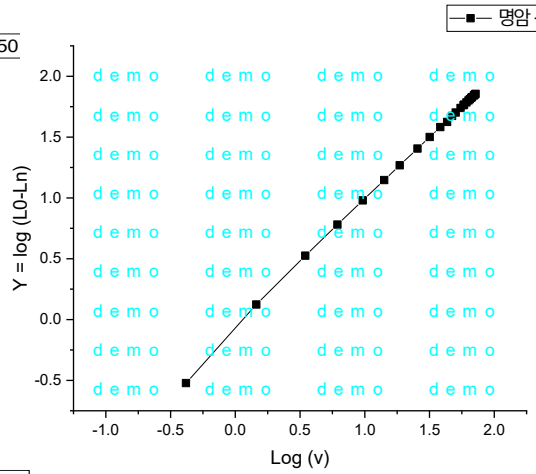
PDP LCD 측정 데이터

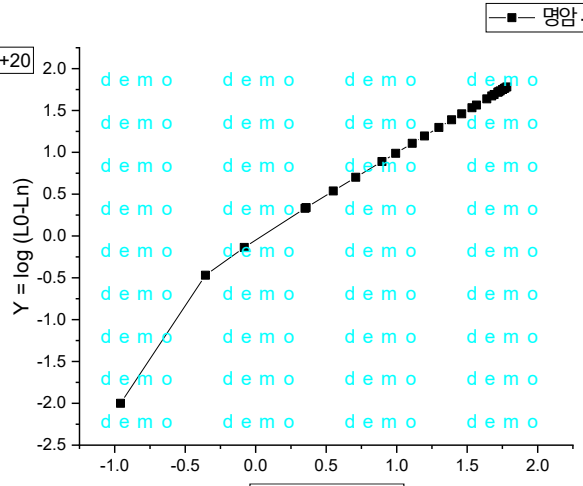


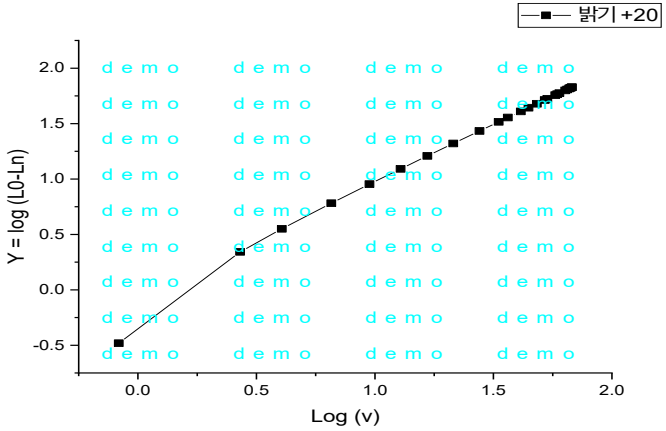


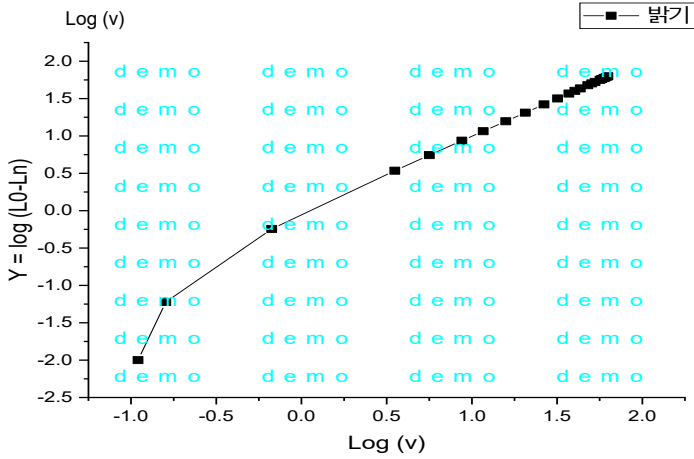
PDP 측정 데이터


명암 80 밝기 50
Y = log (L0-Ln)
Log (v)
demo

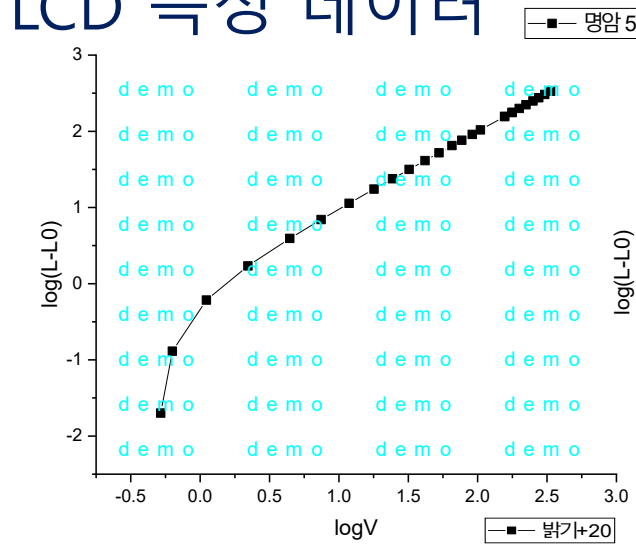

명암 +20
Y = log (L0-Ln)
Log (v)
demo

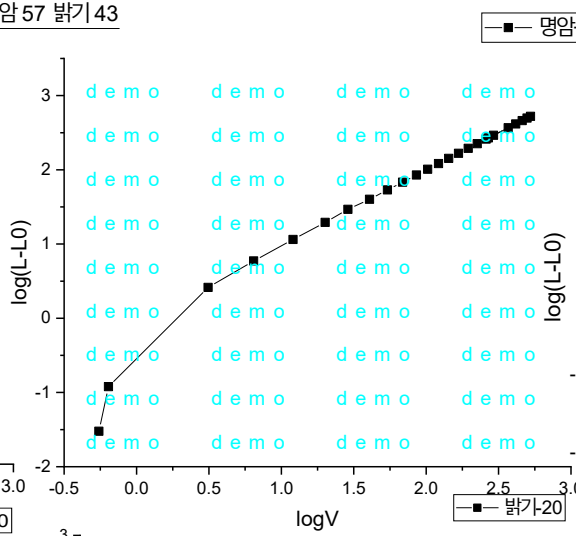

명암 -20
Y = log (L0-Ln)
demo

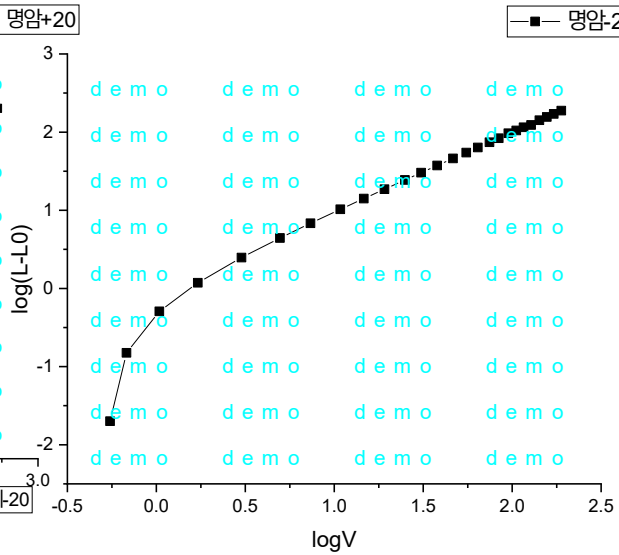

밝기 +20
Y = log (L0-Ln)
Log (v)
demo

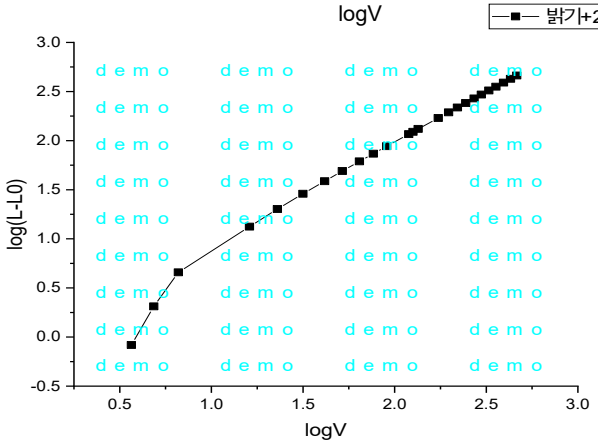

밝기 -20
Y = log (L0-Ln)
Log (v)
demo

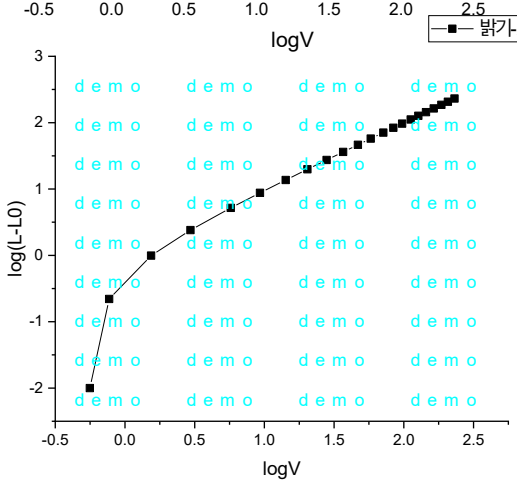
LCD 측정 데이터


명암 57 밝기 43
log(L-L0)
logV
demo


명암+20
log(L-L0)
logV
demo


명암-20
log(L-L0)
logV
demo


밝기+20
log(L-L0)
logV
demo


밝기20
log(L-L0)
logV
demo

Gamma 평균값

	표준	명암+20	명암-20	밝기+20	밝기-20	평균	표준편차
PDP	1.0004	1.0003	1.0004	1.0004	1.0003	1.0004	0.0001
LCD	1.0002	1.0001	1.0005	1.0010	1.0004	1.0004	0.0004

$$\gamma = \frac{\Delta y}{\Delta x} = \frac{\Delta \log(L - L_0)}{\Delta \log(V)}$$

고찰 및 검토

1. ORIGIN 프로그램을 사용하여 그래프를 작성하는데 더 편하기는 했지만 처음 쓰는 프로그램 이여서 약간의 미숙한 점이 남아있었다.

2. 그림 2.5.7을 한번에 나타냈어야 하는데 프로그램의 미숙한 사용으로 하나의 그래프로 나타내지 못하여서 다음에 프로그램을 사용하게 되면 자료를 더 찾아보고 완벽하게 할 수 있도록 해야겠다.

3. 그래프에서 Gray Level 에서 측정 휘도값이 같은 부분이 있는데 그 부분은 전력소모 때문에 saturation 될 수도 있어서 같은 값으로 측정될 수 있기 때문에 log V와 log L-L0값이 음수로 나온다.